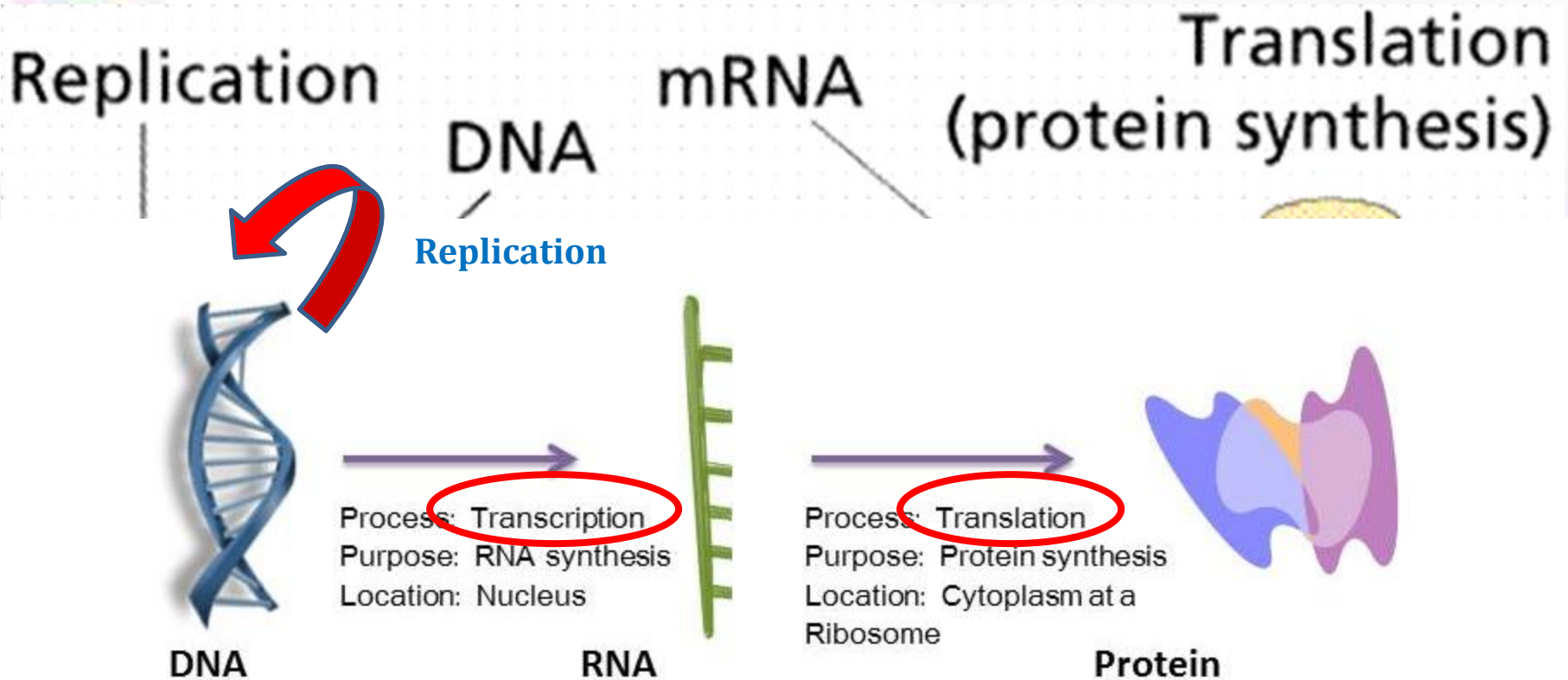


Central Dogma of Molecular Biology

Lecture 10-11

The Central Dogma of Molecular Biology

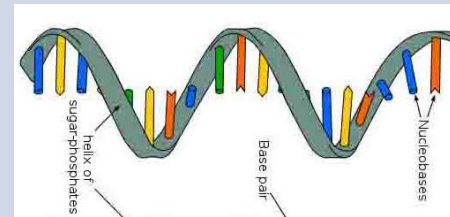


The “Central Dogma”

DNA



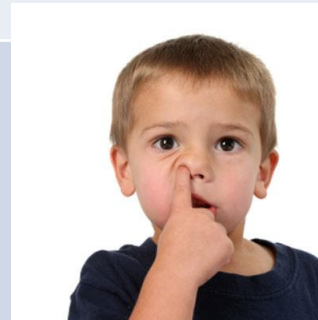
RNA

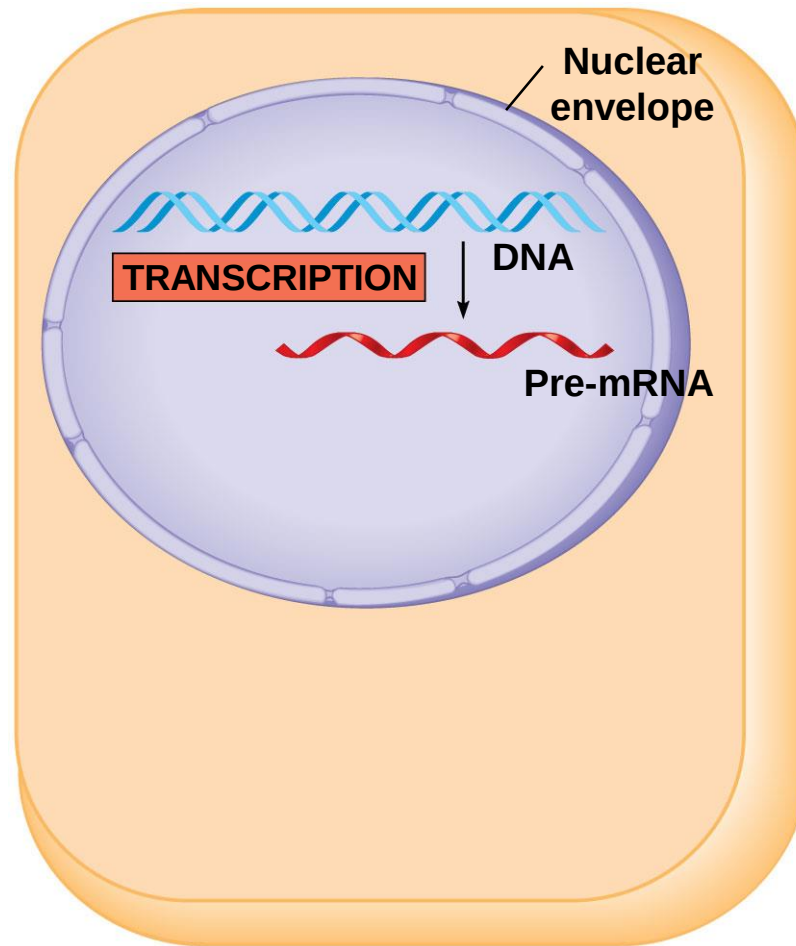


proteins

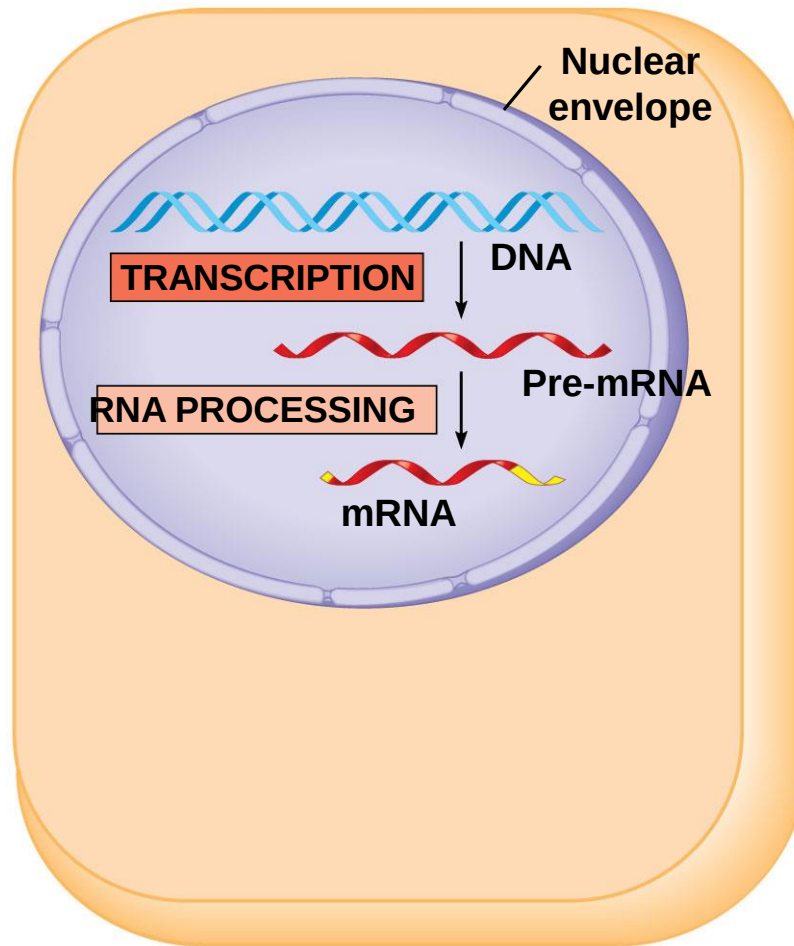


you

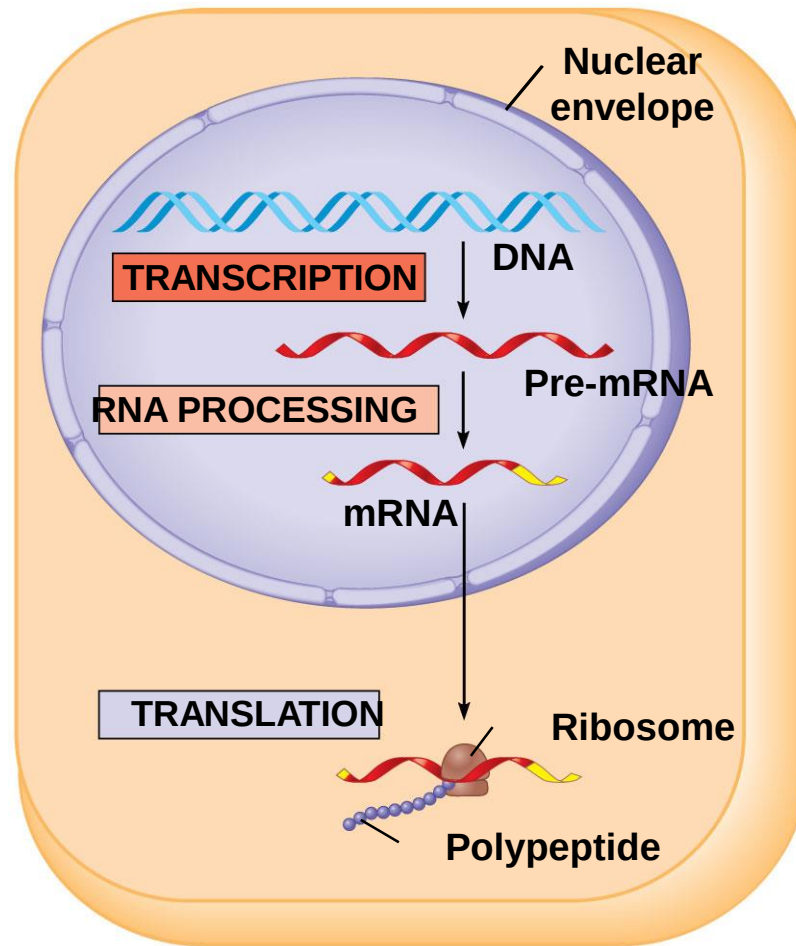




(b) Eukaryotic cell



(b) Eukaryotic cell

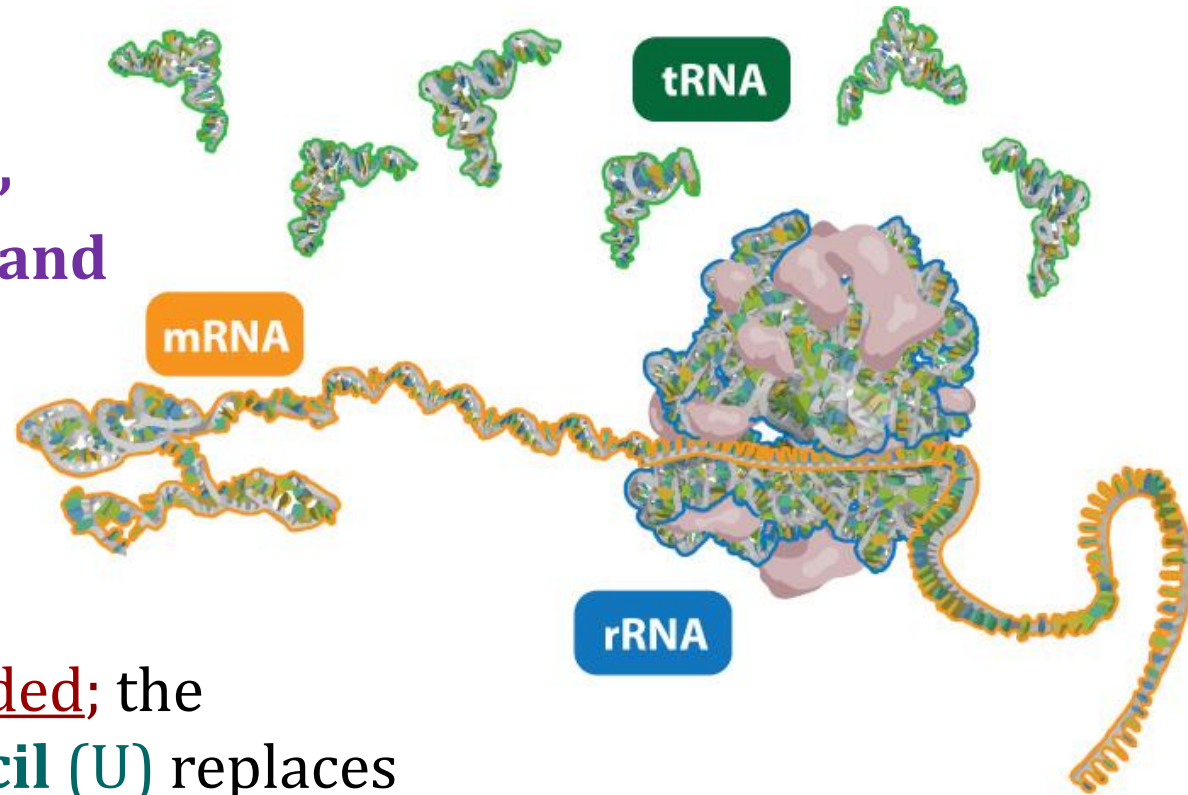


(b) Eukaryotic cell

The RNA

Three major classes of RNA:

- Messenger (mRNA),
 - Ribosomal (rRNA), and
 - Transfer (tRNA)
-
- RNA is a single stranded; the pyrimidine base **uracil** (U) replaces **thymine** and **ribose** sugar replaces **deoxyribose**.



Types of RNA

mRNA



Processing



Messenger RNA (mRNA):
responsible for carrying
information from genes
for the production of
polypeptides in the cytoplasm.

tRNA



+ Amino acids



Transport RNA (tRNA):
responsible for transporting
amino acids to ribosomes
for protein synthesis.

rRNA



+ Ribosomal proteins

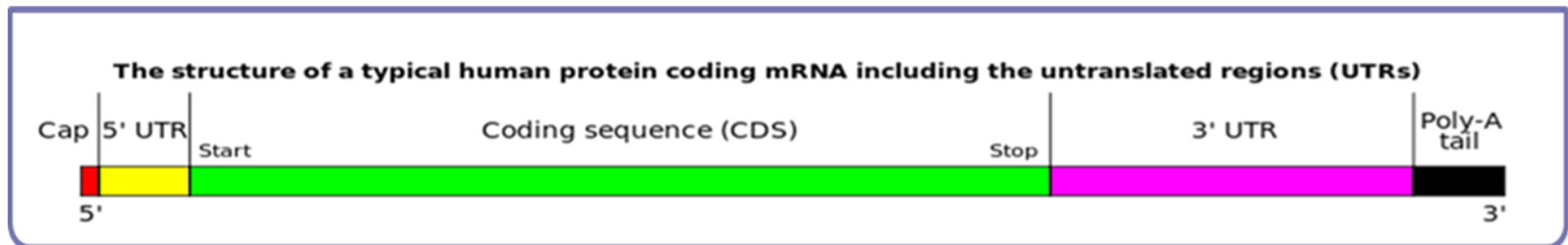


Ribosome

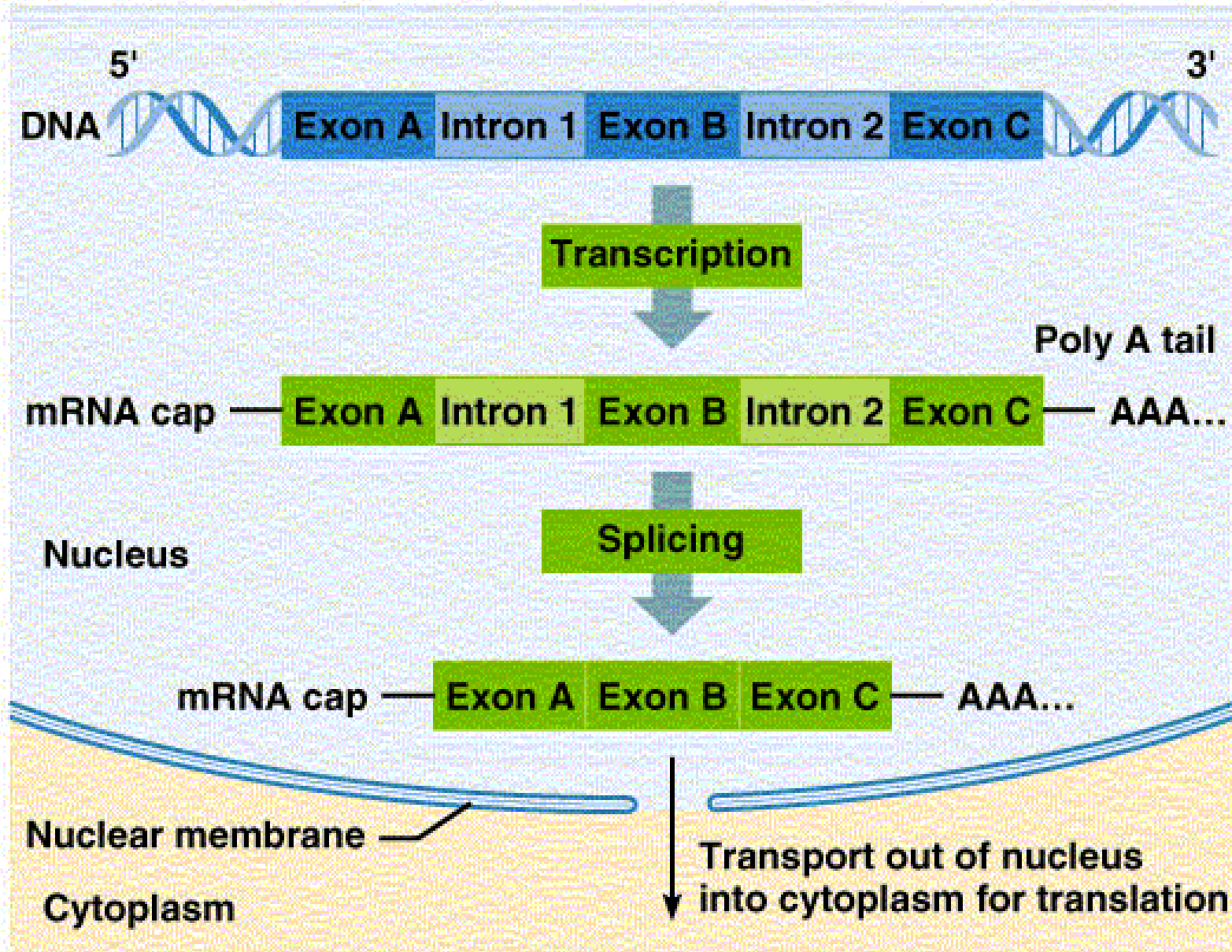
Ribosomal RNA (rRNA):
associated with ribosomal
proteins form ribosomes.

Messenger RNA/ mRNA

- Transcripts of structural genes.
- Encode all the information necessary for the synthesis of a polypeptide of protein.
- The **5' terminus** is **capped** by **7 methylguanosine triphosphate**.
- **Synthesis of the poly (A)** tail involves cleavage of its 3' end and then the addition of about 200 adenine residues.
- Intermediate carrier of genetic information; deliver genetic information to the cytoplasm.

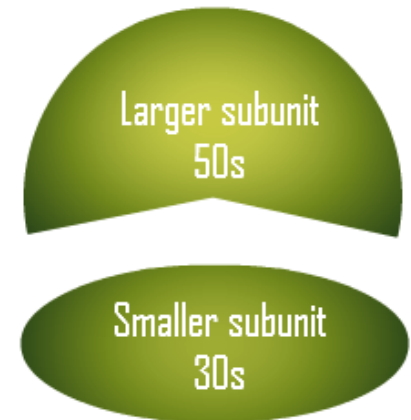
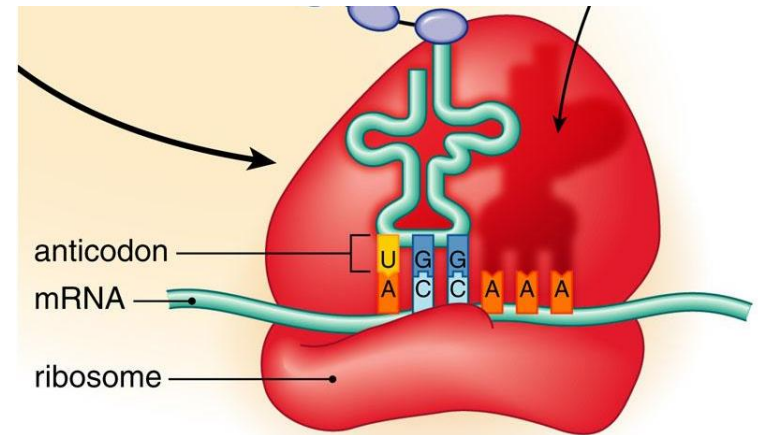


Messenger RNA processing



Ribosomal RNA/ rRNA

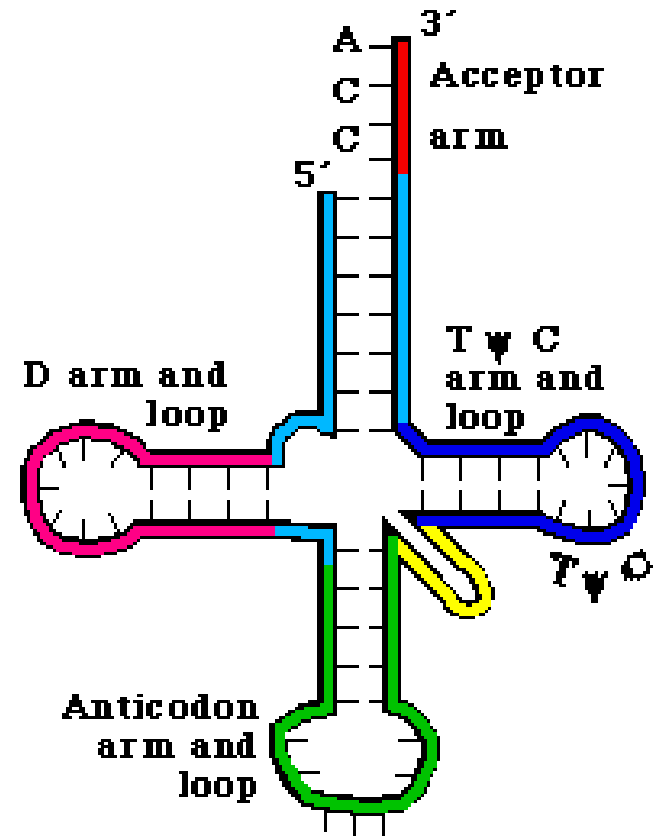
- Ribosome; factory for protein synthesis; composed of ribosomal RNA and ribosomal proteins
- rRNA provides a mechanism for decoding mRNA into amino acids.



Structure of ribosomal subunit

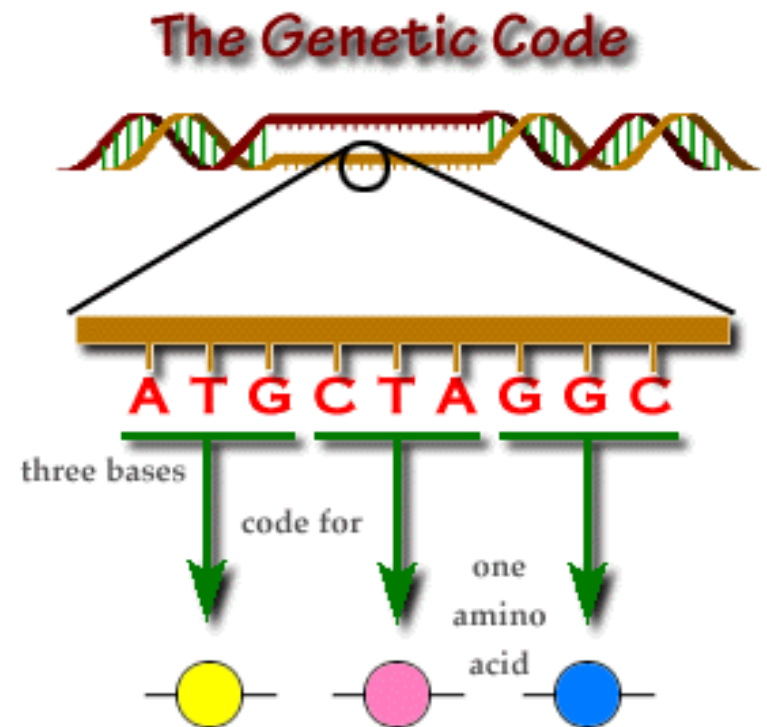
Transfer RNA/ tRNA

- All the tRNAs share a common secondary structure resembles a cloverleaf: They have four base-paired stems defining three stem-loops (the D loop, anticodon loop, and T loop) and **the acceptor stem**.
- tRNA carry correct amino acids to their position along the mRNA template to be added to the growing polypeptide chain.



What is the genetic code?

- **Messenger RNA (mRNA)** carries DNA's protein-building information to ribosomes for translation
- mRNA's genetic message is written in **codons**
 - Sets of three nucleotides along mRNA strand



Codons

- **Codons** specify different amino acids
 - A few codon signals stop translation
- **Sixty-four** codons constitute a highly conserved **genetic code**

first base ▼	second base				third base ▼
U	U	C	A	G	U C A G
UUU } phe	UCU } ser	UAU } tyr	UGU } cys		
UUC }	UCC }	UAC }	UGC }		
UUA } leu	UCA }	UAA STOP	UGA STOP		
UUG }	UCG }	UAG STOP	UGG trp		
C	U	C	A	G	U C A G
CUU } leu	CCU } pro	CAU } his	CGU } arg		
CUC }	CCC }	CAC }	CGC }		
CUA }	CCA }	CAA } gln	CGA }		
CUG }	CCG }	CAG }	CGG }		
A	U	C	A	G	U C A G
AUU } ile	ACU } thr	AAU } asn	AGU } ser		
AUC }	ACC }	AAC }	AGC }		
AUA }	ACA }	AAA } lys	AGA } arg		
AUG met	ACG }	AAG }	AGG }		
G	U	C	A	G	U C A G
GUU } val	GCU } ala	GAU } asp	GGU } gly		
GUC }	GCC }	GAC }	GGC }		
GUA }	GCA }	GAA } glu	GGA }		
GUG }	GCG }	GAG }	GGG }		

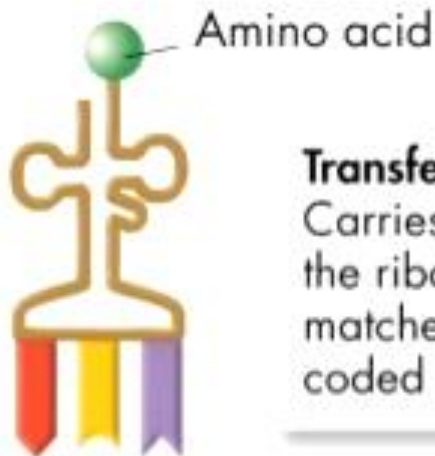
Roles of RNA in Translation

All three major forms of RNA—mRNA, tRNA, and rRNA—are involved in the process of translation.



Messenger RNA

Carries instructions for polypeptide synthesis from nucleus to ribosomes in the cytoplasm.



Transfer RNA

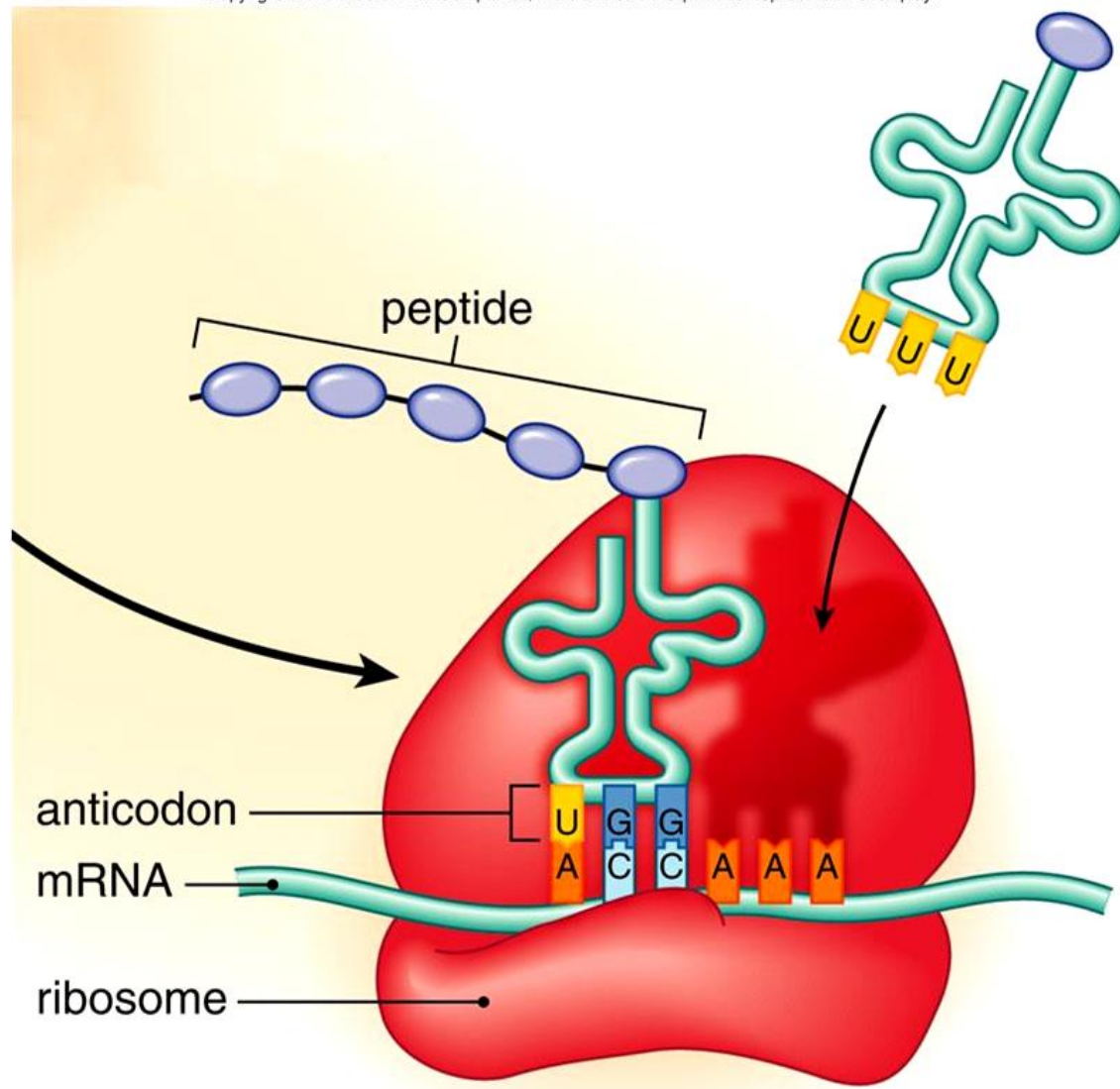
Carries amino acids to the ribosome and matches them to the coded mRNA message.



Ribosome

Ribosomal RNA

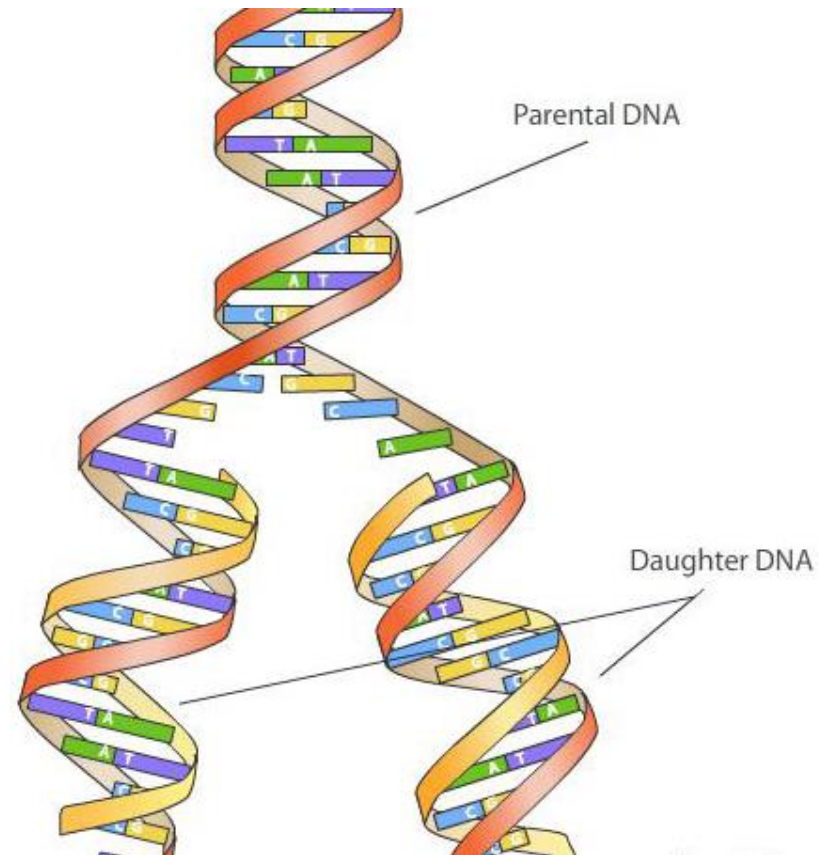
Forms an important part of both subunits of the ribosome.



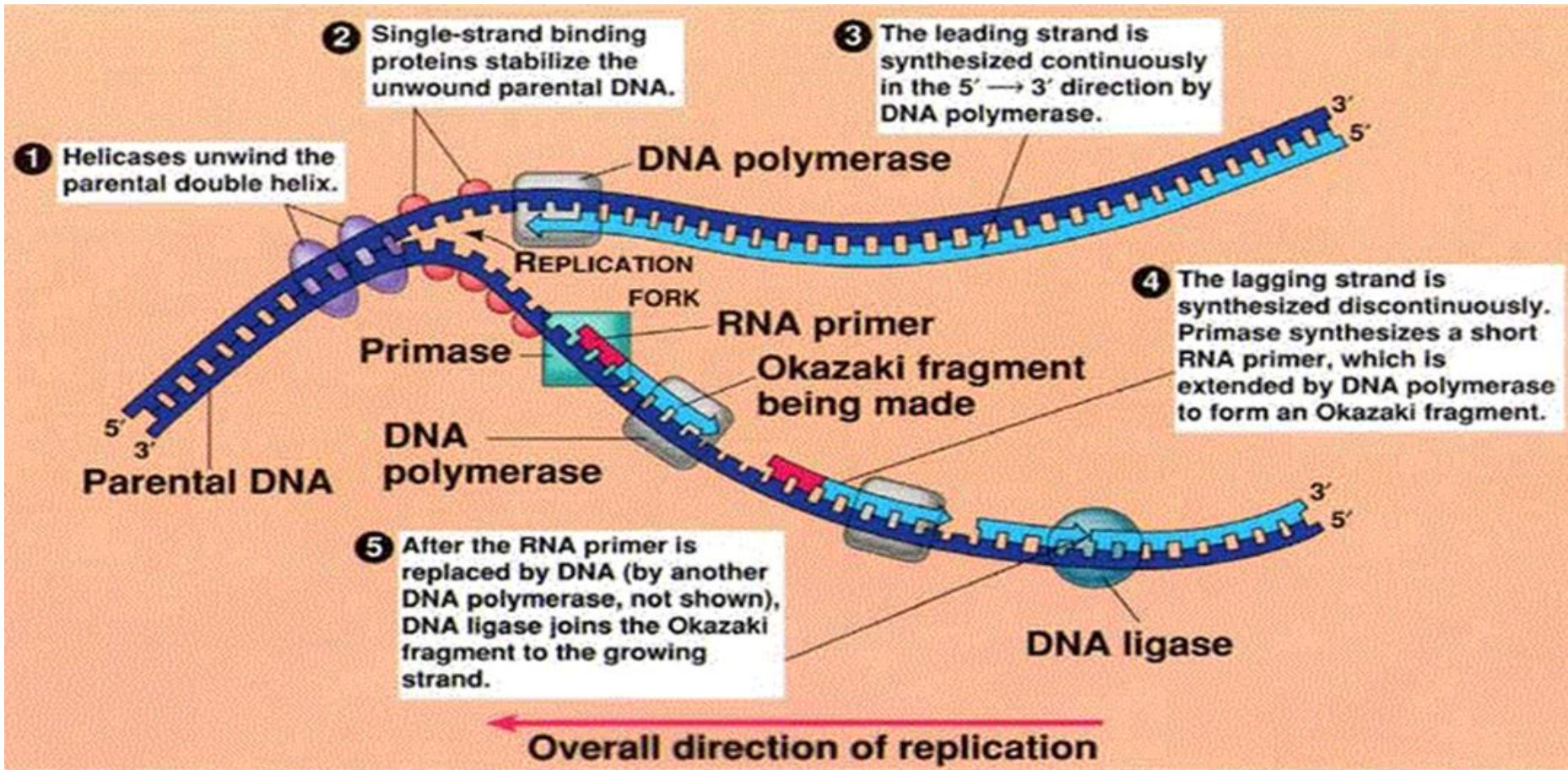
tRNA-amino acid at ribosome

DNA Replication

- The DNA duplication.
- The transfer the genetic information from a parent to a daughter cell.
- The DNA base sequences are precisely copied.

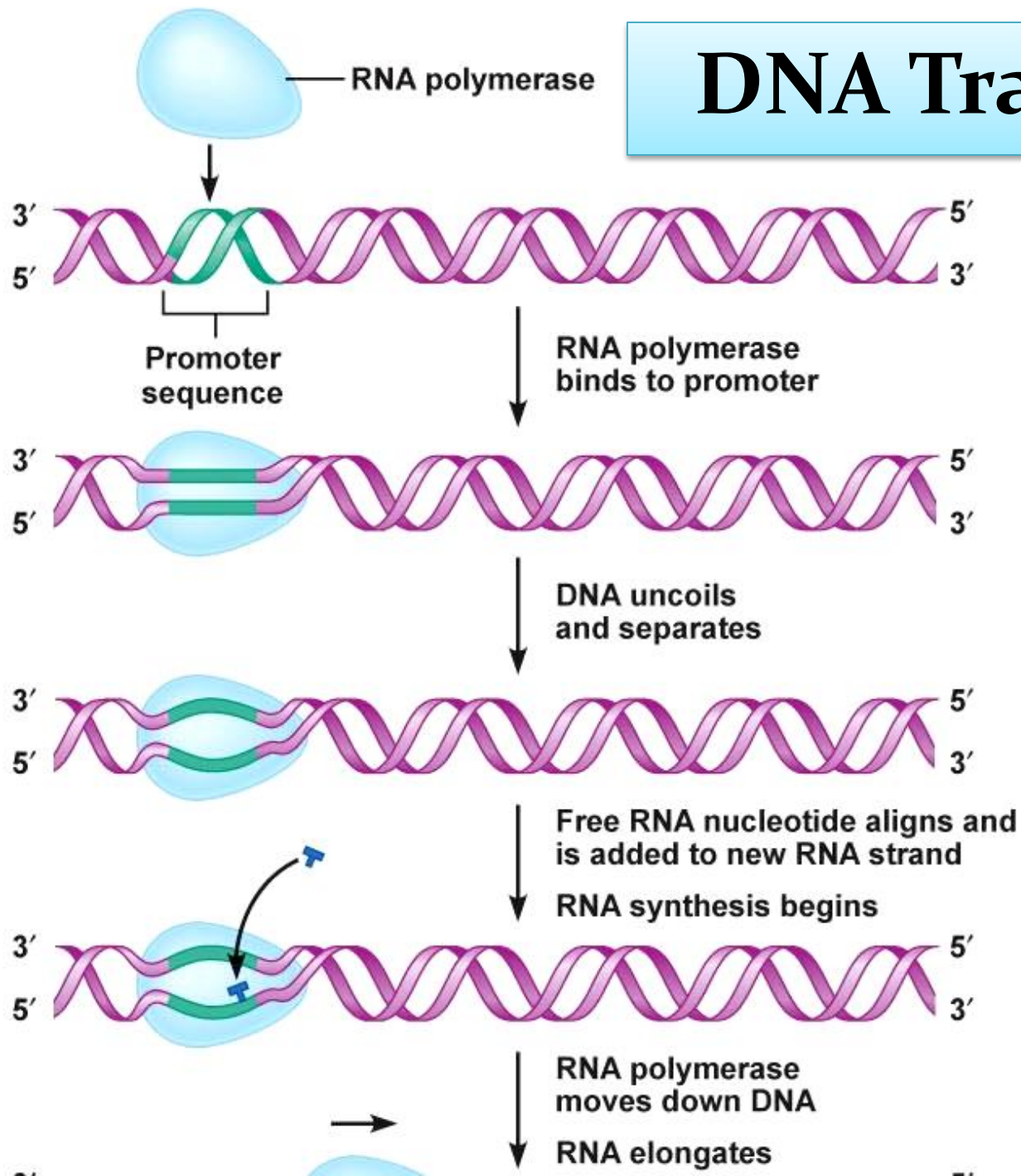


Summary of DNA Replication



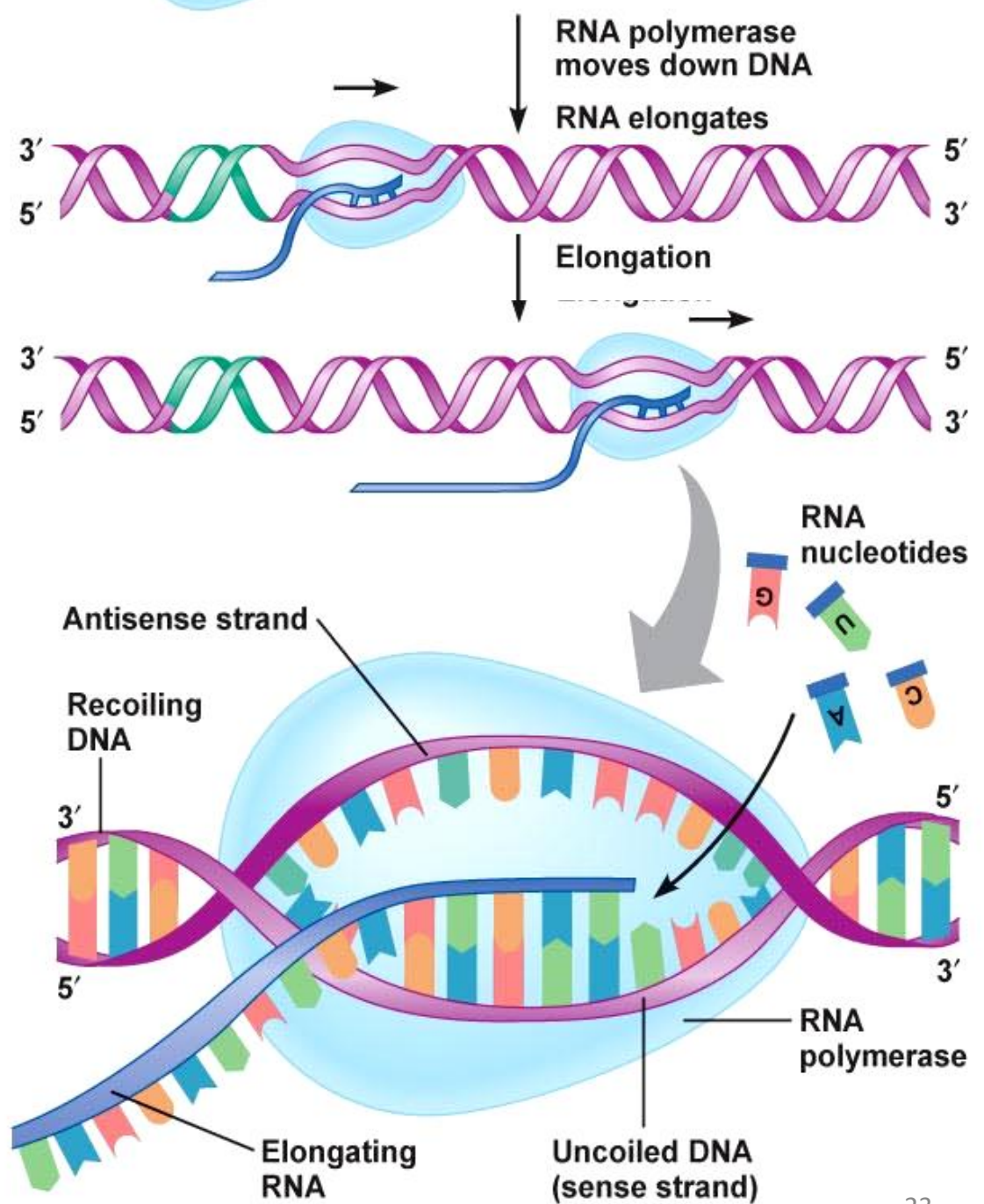
DNA are synthesized from 5' to 3' direction

DNA Transcription

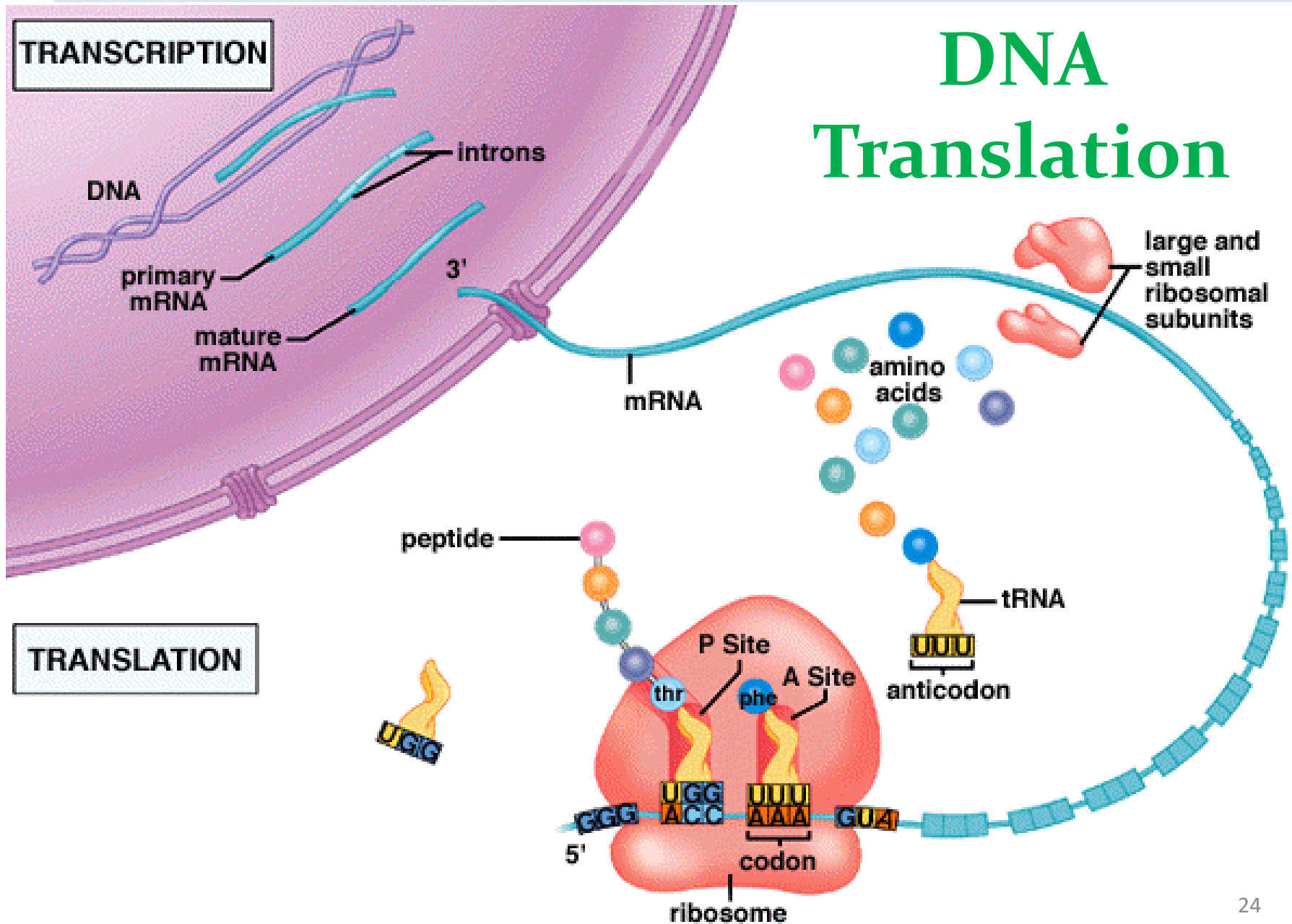


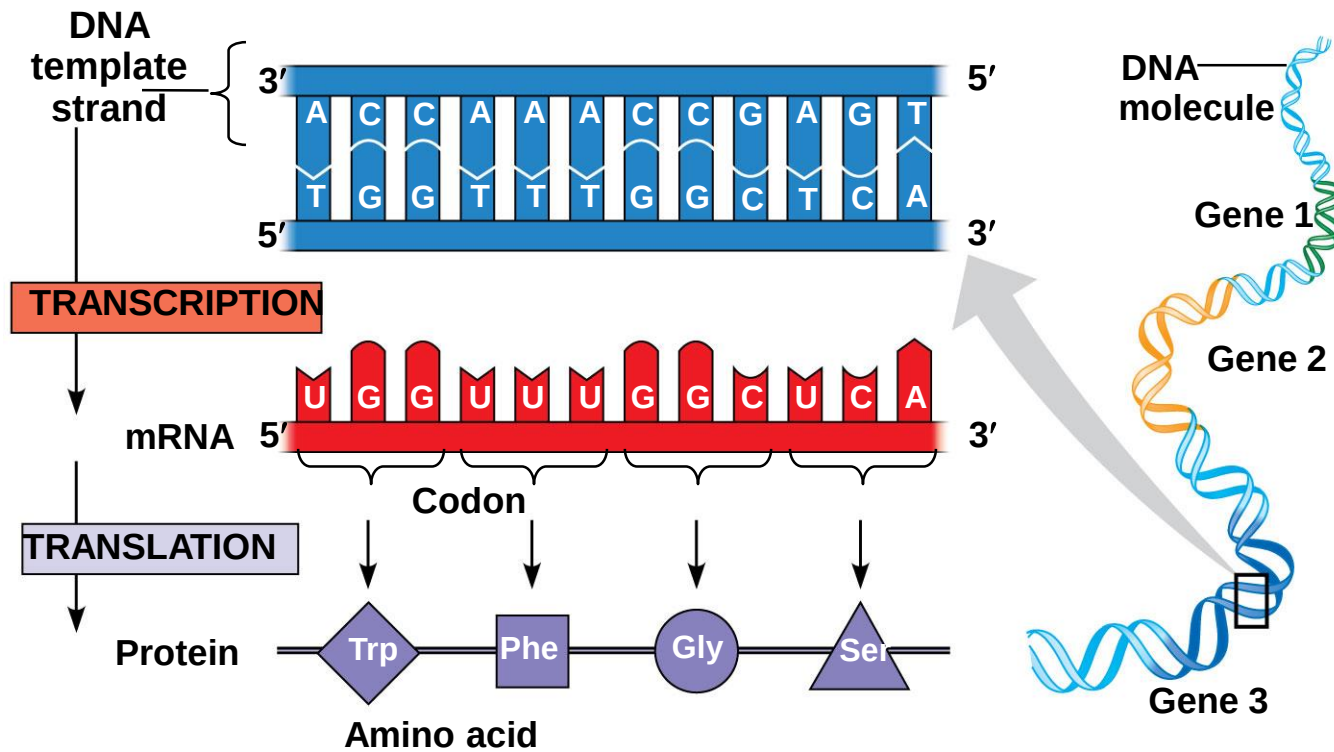
mRNA
intermediate is
transcribed from
one of the strands
of the DNA
molecule

DNA Transcription (Cont...)



Summary of Protein Synthesis



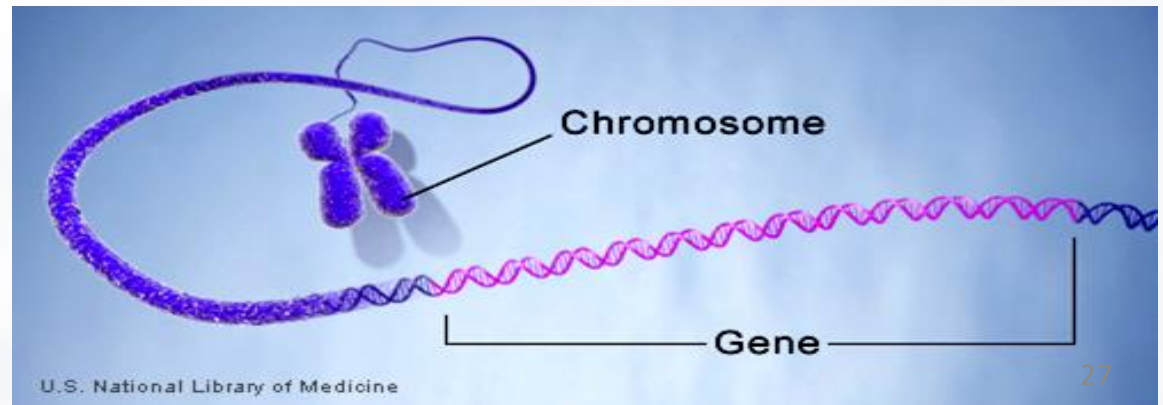
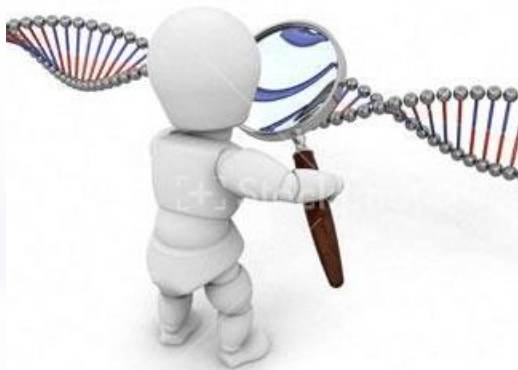


Comparison between translation and transcription

	Transcription	Translation
Purpose	To make RNA copies of individual genes	To synthesize proteins which are used for cellular functions
Definition	Transcription is the synthesis of RNA from a DNA template.	Translation is the synthesis of a protein from an mRNA template. i.e., translating the DNA genetic information carried by mRNA to synthesize proteins. This is the second step of gene expression.
Products	mRNA	Proteins
Location	Nucleus	Cytoplasm

Genes

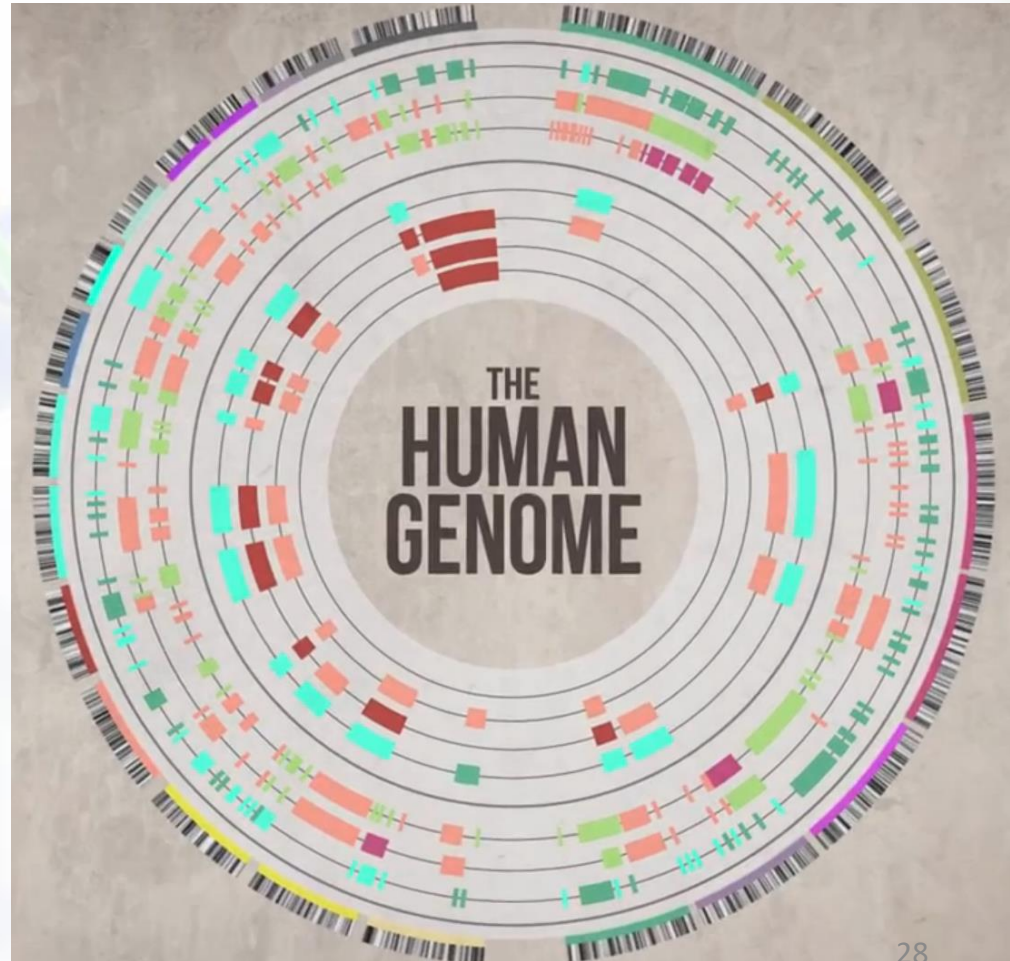
- * The DNA in each chromosome constitutes many genes.**
- * The DNA also contains large sequences that do not code for any protein and their function is not known.**
- * The gene of the coding region encodes instructions that allow a cell to produce a specific protein or enzyme.**
- There are nearly 50,000 to 100,000 genes with each being made up of hundreds of thousands of chemical bases.**



The Human Genome

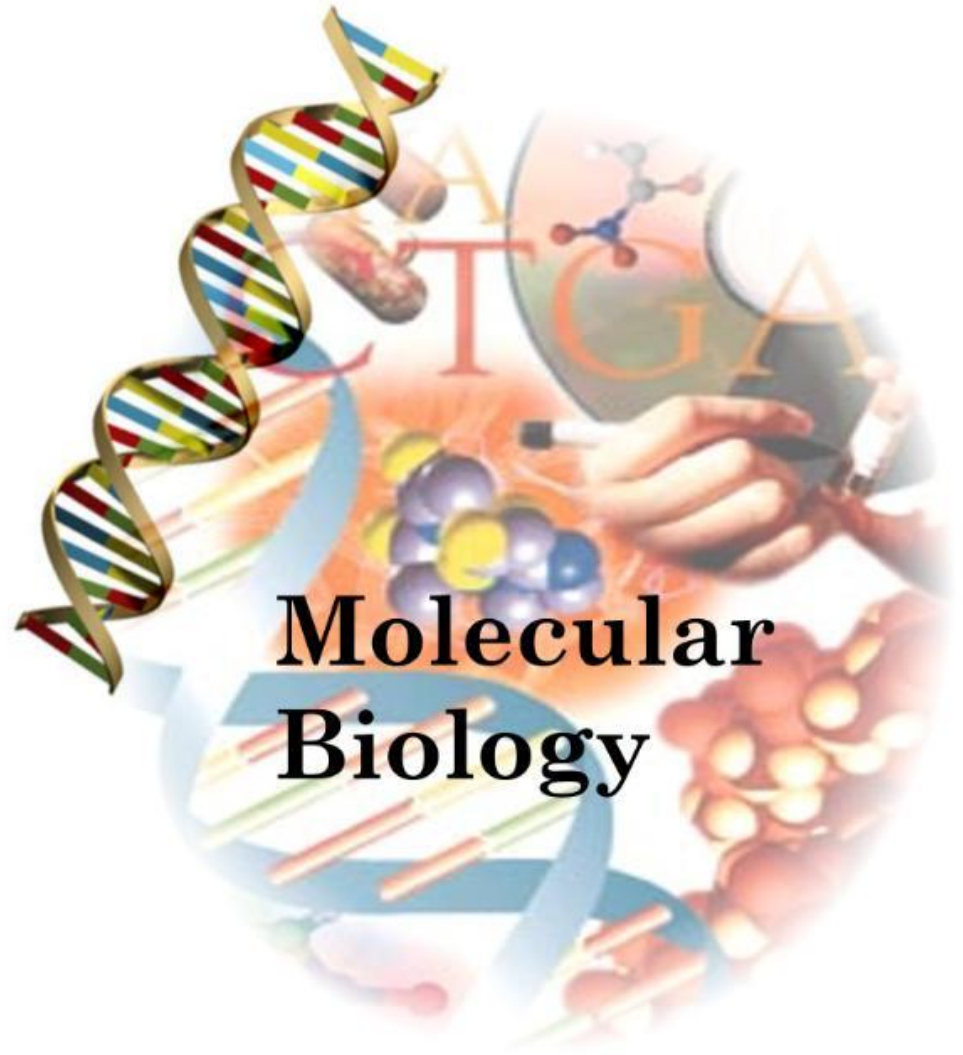
All the DNA in the cell makes up the human genome.

There are about 20,000 important genes located on one of the 23 chromosome pairs found in the nucleus and/or on long strands of DNA located in the mitochondria.



Application of Molecular Biology

- **Research**
- **Diagnosis**
- **Transplantation**
- **Paternity**
- **Forensic analysis**
- **Gene therapy**
- **Drug Design**



Thank You